

AMNA FIAZ

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience building production AI systems, document-processing platforms, agentic RAG applications, data pipelines, backend services, and automation tools. Experienced with Python, FastAPI, Django, PostgreSQL, Redis, Docker, AWS, and GCP. Worked on projects involving LLMs, RAG, OCR, PDF verification, web automation, data engineering, and multi-agent workflows, with a strong focus on building reliable systems that solve real business problems.

TECHNICAL SKILLS

AI & LLM Systems	RAG, LangChain, LangGraph, LLM Agents, Tool Calling, FAISS, Groq, HuggingFace, Prompt Engineering, LLM Evaluation
Python & Backend	Python, FastAPI, Django, REST APIs, GraphQL, Pydantic, SQLAlchemy, Uvicorn, async/await, Celery
Data & Databases	PostgreSQL, MySQL, BigQuery, SQL, Pandas, NumPy, schema design, query optimization, data validation
Document AI & Automation	PyMuPDF, PaddleOCR, Playwright, PDF Parsing, OCR, Regex Extraction, Template Matching, n8n
Cloud & DevOps	Docker, Docker Compose, Git, GitHub Actions, Linux, AWS EC2, GCP, Cloud Composer, Redis, SSH

WORK EXPERIENCE

Black Lion Innovations Group — Data Automation Scientist

Mar 2026 – Present

- Built and deployed FastAPI services and automated data pipelines on GCP and AWS EC2, using Docker, PostgreSQL, BigQuery, background jobs, and production monitoring.
- Built an AI assistant for women's hormonal health using RAG and agent-based workflows, allowing the system to search trusted knowledge, keep conversation context, and give more relevant answers to users.
- Developed Playwright-based collection pipelines for Spotify artist, playlist, and web data, processing more than 50,000 records in a run with background workers and isolated subprocesses.
- Added automated checks for missing values, duplicates, incorrect schemas, and inconsistent records across more than 15 data sources, cutting down manual data-review work.

LUMS – BIRL — AI Engineer

May 2025 – Feb 2026

- Built Python ETL pipelines and FastAPI services for cleaning, transforming, combining, and reporting data from 8 clinical datasets.
- Developed SQL, Pandas, and NumPy workflows for feature engineering, data validation, and anomaly checks used by downstream research teams.
- Improved PostgreSQL queries and processing logic, reducing end-to-end data-processing time by around 40%.
- Parallelized independent processing steps and simplified pipeline logic so larger datasets could be handled more reliably.

LUMS – Prof. Abdul Basit — Freelance AI/Backend Engineer

Project Based

- Built an automated document-verification platform for Spanish energy-efficiency audit claims, covering 11 document types and roughly 40–50 fields per document.
- Developed PDF parsing, OCR, field extraction, validation, and cross-document checks to compare names, IDs, addresses, prices, measurements, signatures, and audit data.
- Generated marked-up PDFs with green, red, and amber boxes directly on the original pages so reviewers could quickly see correct, incorrect, and unverified information.

PROJECTS

Automated Document Verification Platform – Spanish Energy-Efficiency Audits (CAE)

- Built a document-checking platform for Spanish energy-efficiency claims that reads 11 different document types and checks around 40–50 fields from each document against the recorded audit information.
- Combined PyMuPDF, PaddleOCR, PostgreSQL, and rule-based validation to check DNI/NIE/CIF numbers, cadastral references, addresses, measurements, VAT calculations, prices, signatures, and values repeated across several documents.
- Created annotated PDFs that place green, red, and amber boxes over the exact fields on the page, with explanations showing reviewers what matched, what was wrong, and what still needed manual checking.

Women's Hormonal Health Agentic RAG Platform

- Built an AI assistant for women's hormonal-health questions using RAG so answers could be based on retrieved knowledge instead of relying only on the language model.
- Added agent-based workflows for handling different user questions, retrieving relevant information, keeping conversation context, and passing tasks between different parts of the system.
- Developed the backend and AI workflow as part of work at Black Lion Innovations Group, with a focus on making the assistant practical, easy to extend, and reliable for repeated conversations.

Spotify Data Intelligence Platform | GitHub

- Built a FastAPI and Playwright service for collecting Spotify artist, playlist, and related web data through automated browser jobs.
- Added Pydantic validation, API-key protection, background subprocess jobs, and endpoints for status, logs, cancellation, health checks, and downloads.
- Added CSV exports and optional BigQuery integration so collected data could move directly into analytics and reporting workflows.

WhatsApp Visa Processing AI Agent

- Built an automated visa-processing workflow using n8n and the Twilio WhatsApp API so users could send documents and information through WhatsApp.
- Used LlamaParse and Groq to read uploaded documents, extract important information, and pass structured results through the workflow.
- Connected Google Sheets, webhooks, and human approval steps so staff could review important cases before the system continued.

Medical RAG Chatbot | GitHub

- Built a document question-answering chatbot that lets users ask questions from uploaded medical PDF documents.
- Used LangChain, FAISS, embeddings, Groq, and HuggingFace models to split documents, retrieve relevant text, and generate answers from the retrieved context.
- Created the user interface in Streamlit and kept API keys and other private configuration outside the application using environment variables.

Agentic Dino Platform | GitHub

- Built an LLM-powered agent platform with LangGraph where agents can call tools, stop for human approval, and continue a workflow from saved checkpoints.
- Added versioned prompts and reusable workflow components so agent behaviour could be changed and tested without rebuilding the complete system.
- Added evaluation and MLOps workflows using MLflow, LLM-as-a-judge scoring, regression checks, source-controlled datasets, classical ML,

and Databricks deployment.

Spotlytics – Spotify Analytics Dashboard | GitHub

- Built a Spotify analytics dashboard using Django and GraphQL for querying and presenting structured music analytics data.
- Used PostgreSQL to store the application and analytics data needed by the backend and dashboard.
- Added Celery and Redis for background processing so longer data jobs could run without slowing down normal user requests.

COURSES & CERTIFICATIONS

- **Agentic Systems with LangGraph** — DataCamp | [View Credential](#)
- **Certificate of Coordination, 3rd International AI Championship** — AIEF/LUMS (Nov 2025) | [View Certificate](#)
- **Joining Data with pandas** — DataCamp (Jul 2026) | [View Credential](#)

EDUCATION

UET Lahore — BSc Computer Engineering

Sep 2021 – Mar 2025

CGPA: 3.58/4.0